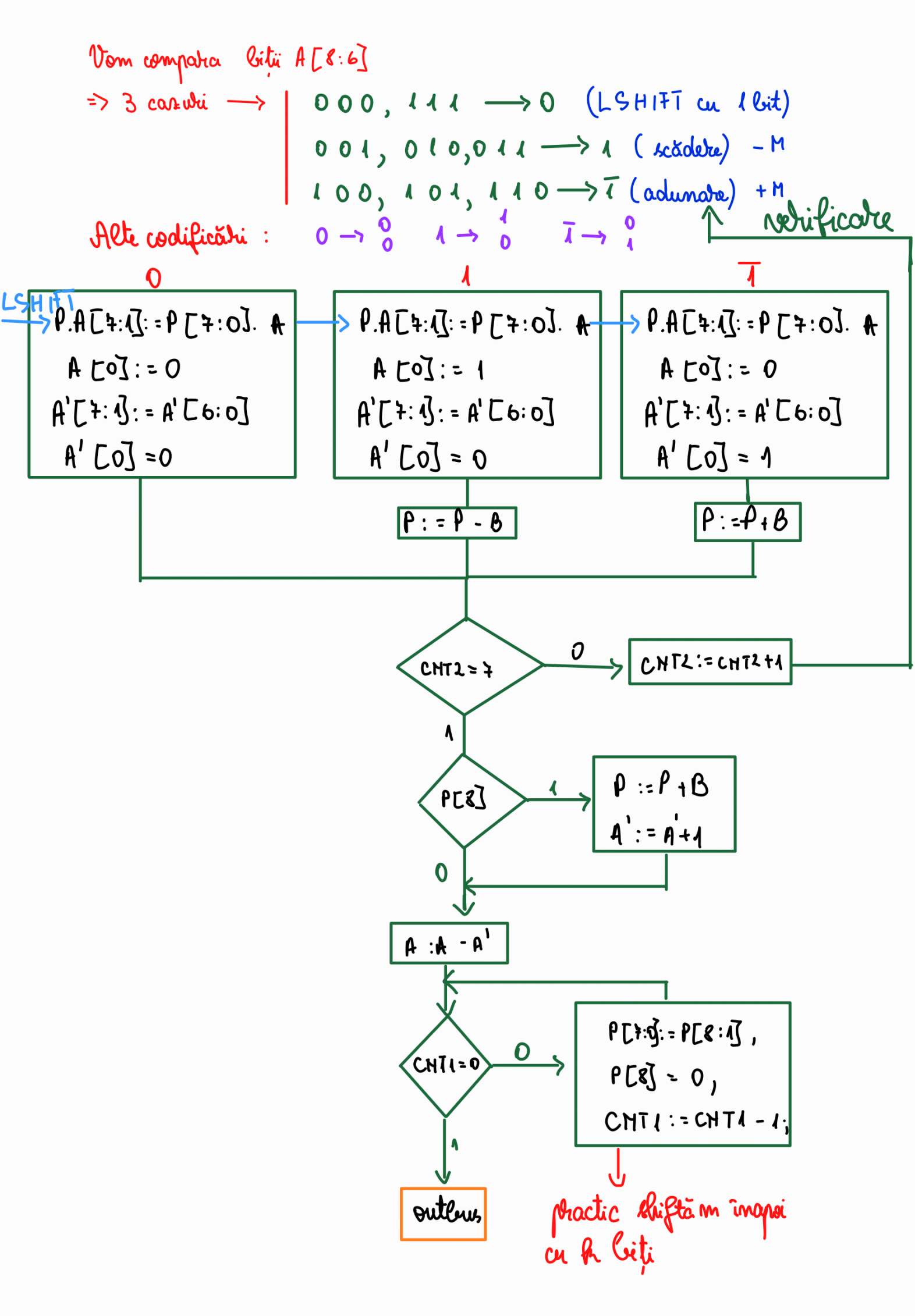

Vom compara biții A[8:6]
=> 3 cazuri →
000, 111 → 0 (LSHIFT cu 1 bit)
001, 010, 011 → 1 (scădere) -M
100, 101, 110 → T (adunare) +M
Alte codificări :
0 → 00
1 → 10
T → 01
verificare
0
1
T
LSHIFT
P.A[7:1] := P[7:0]. A
A[0] := 0
A'[7:1] := A'[6:0]
A'[0] := 0
P.A[7:1] := P[7:0]. A
A[0] := 1
A'[7:1] := A'[6:0]
A'[0] := 0
P.A[7:1] := P[7:0]. A
A[0] := 0
A'[7:1] := A'[6:0]
A'[0] := 1
P := P - B
P := P + B
CNT2 = 7
0
CNT2 := CNT2 + 1
1
P[8]
1
P := P + B
A' := A' + 1
0
A := A - A'
CNT1 = 0
0
P[7:0] := P[8:1],
P[8] := 0,
CNT1 := CNT1 - 1;
1
outLeu
practic shiftăm înapoi cu 8 biți

① e pe 15 biți — 128 64 32 16 8 4 2 1

X = 144 = 1001 0000

Y = 5 = 0000 0101

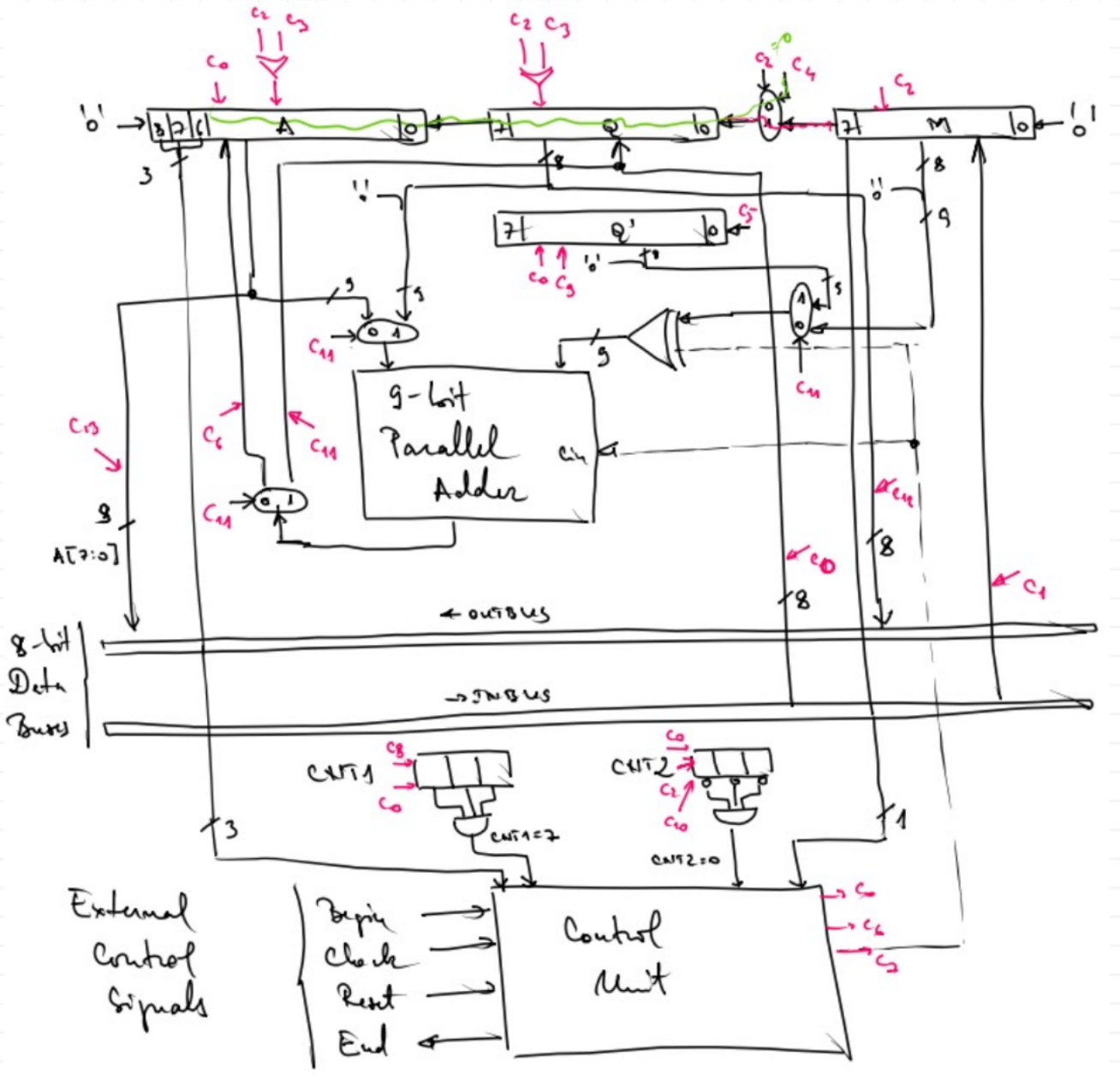
× (începem scrierea din A)

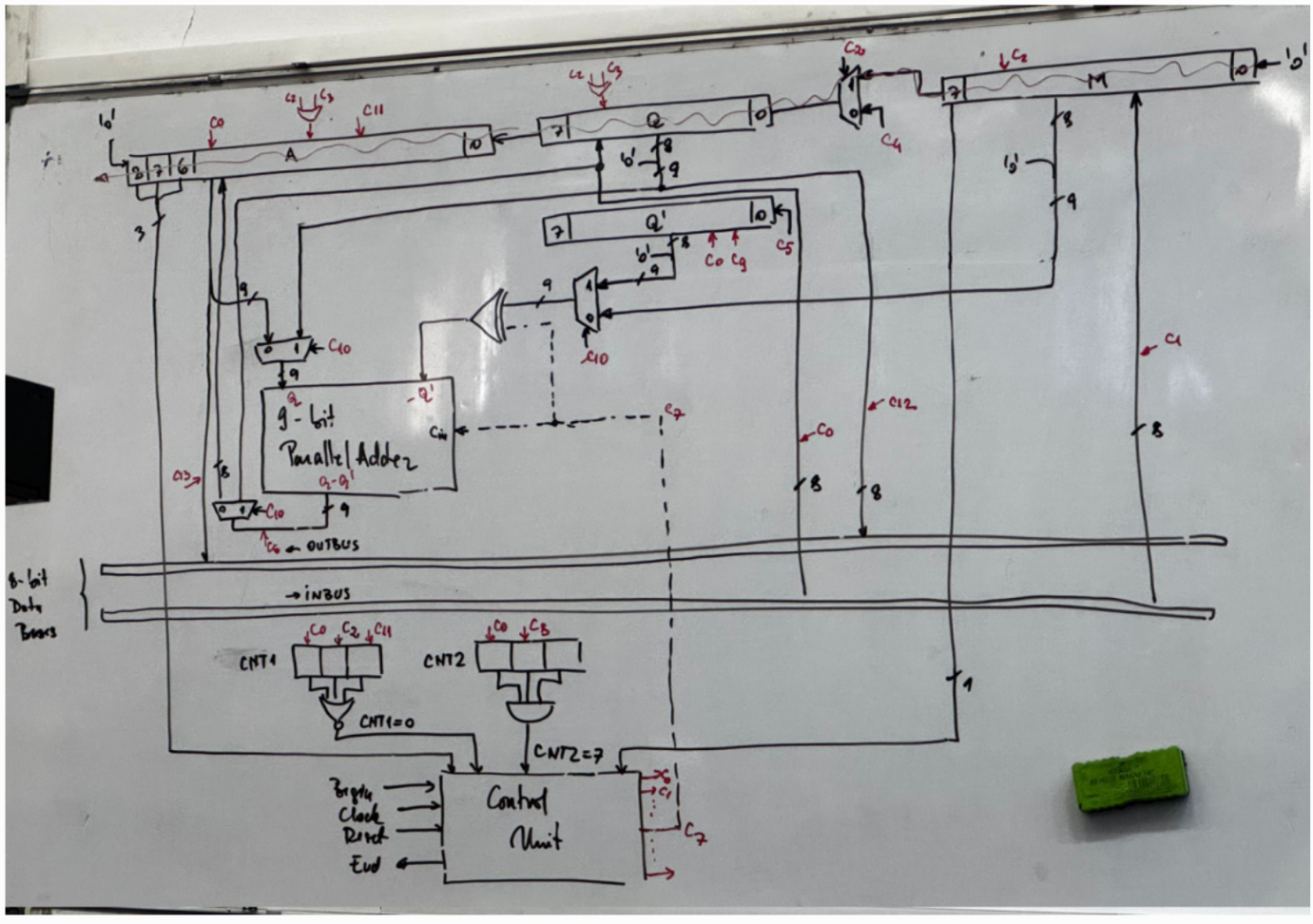
aici se pune Y

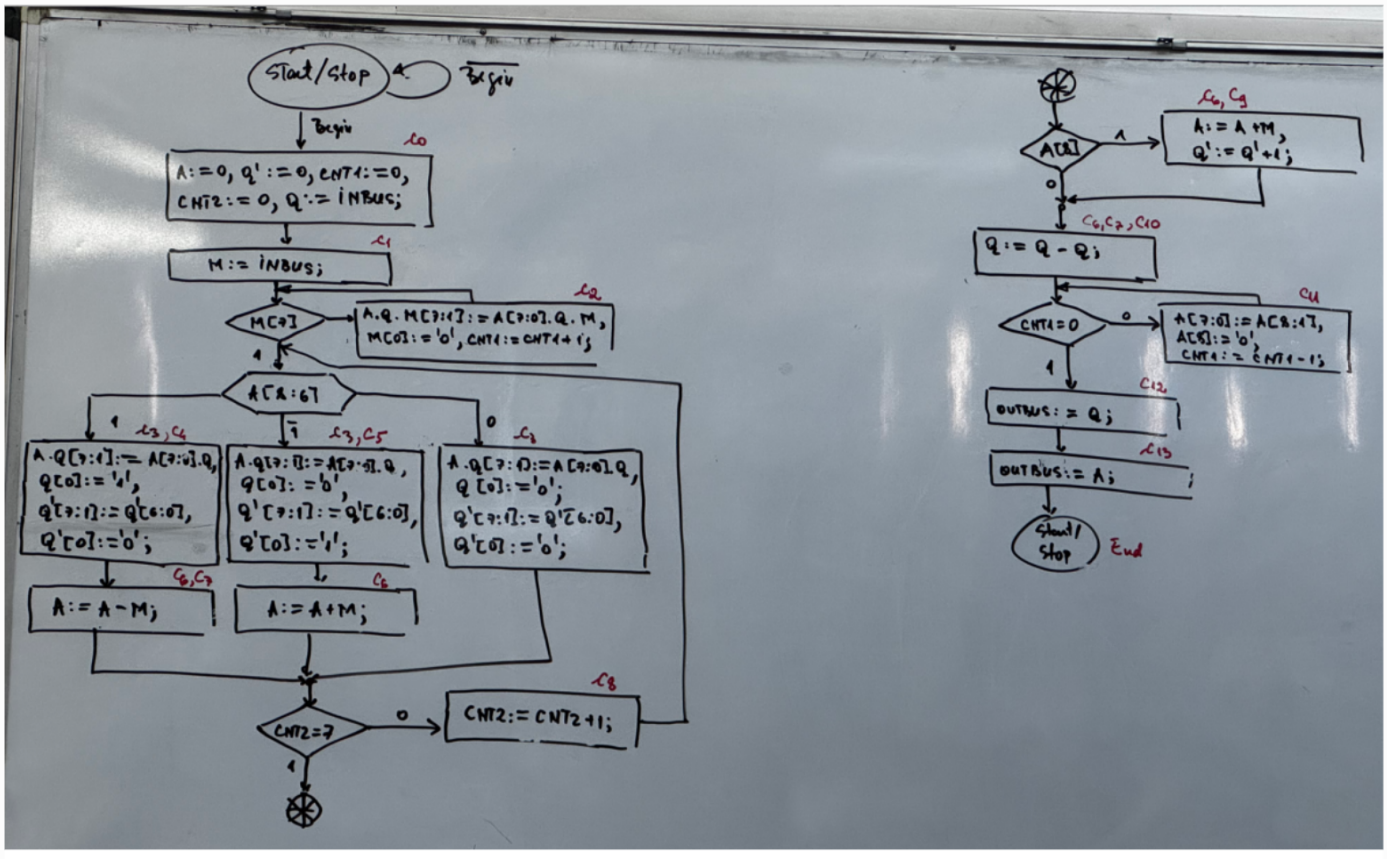
CNT	P	A	B
000	0 0000 0000	1001 0000	0000 0101
	I. mutăm la stânga PAB în funcție de câte leading zeros avem în B		în acest caz $k_1 = 5$
pas inițial ne se crește contor	LSHIFT cu k_1		
	$q_0 = 0$ 0 0001 0010	0000 0000	1010 0000
000	LSHIFT (PA)		
aici comparăm primii 3 biți de la stânga la dreapta	0 0010 0100	0000 0000	adăugați pe baza codului
	$q_1 = 0$		
	LSHIFT (PA)		
001	0 0100 1000	0000 0000	
	$q_2 = 1$	LSHIFT	
010	0 1001 0000	0000 0001	
	⊖ B 0 1010 0000		
cel shiftat ←	1 1111 0000		
	$q_3 = 0$	LSHFT	
011	1 1110 0000	0000 0010	
	$q_4 = 0$	LSHIFT	0000
100	1 1100 0000	0000 0100	
	$q_5 = 0$	LSHIFT	0000
101	1 1000 0000	0000 1000	
	$q_6 = 1$	LSHIFT	0000
110	1 0000 0000	0001 0000	
	⊕ B 0 1010 0000	0001	
	1 1010 0000		
	$q_7 = 1$	LSHIFT	
111	1 0100 0000	0010 0000	
	⊕ 0 1010 0000	0000 0011	⊕ corecție pt că P[3] = 1
	(1) 1110 0000	0000 0100	→ A'

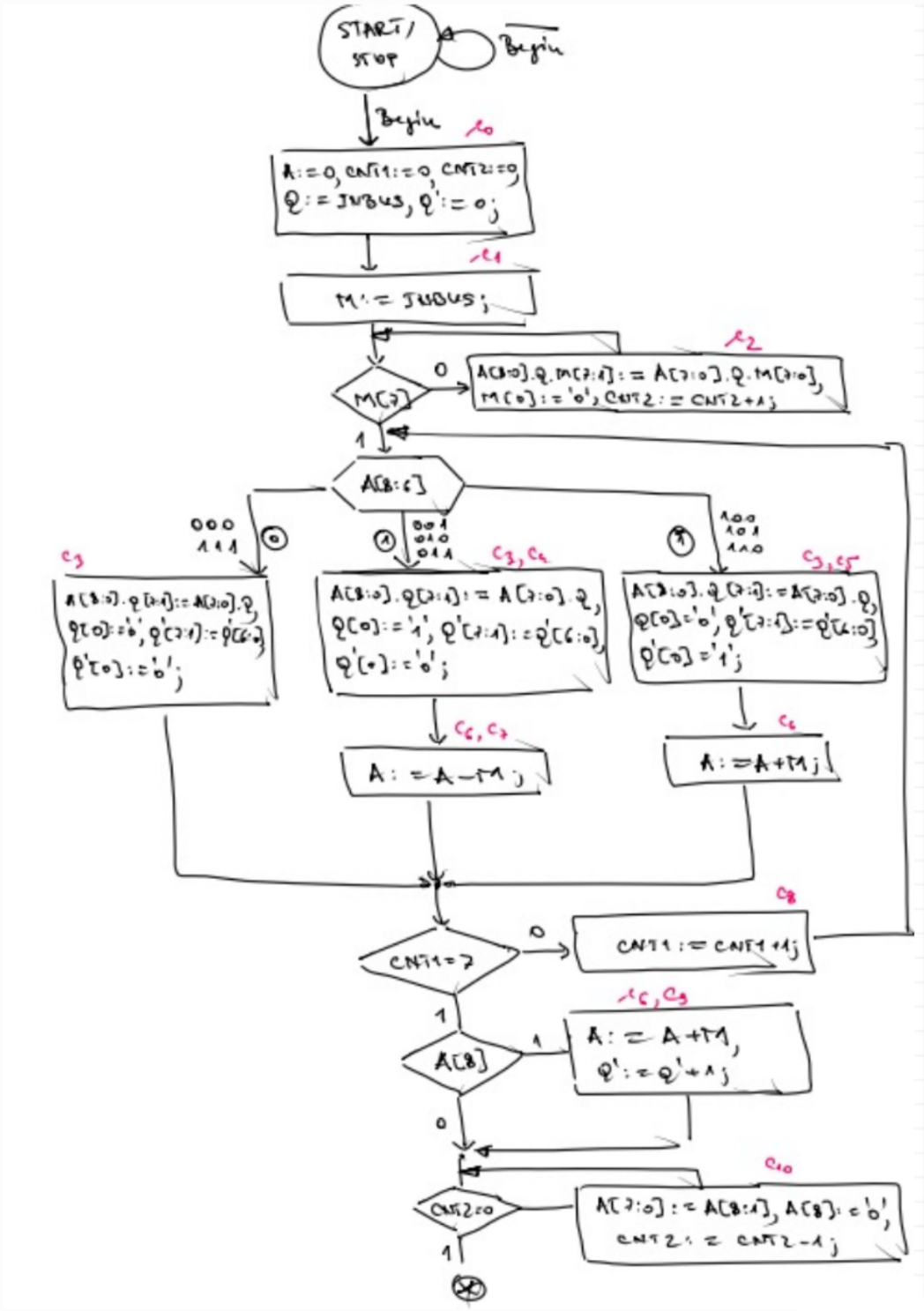
⊕ 0 1010 0000	⊖ 0010 0000
0 1000 0000	0000 0100
RSHIFT cu 4 biți ⑤	0001 1100
0 0000 0100	câtul
restul	4+8+16=28
4	

$28 \cdot 5 + 4 = 140 + 4 = 144$ ✓

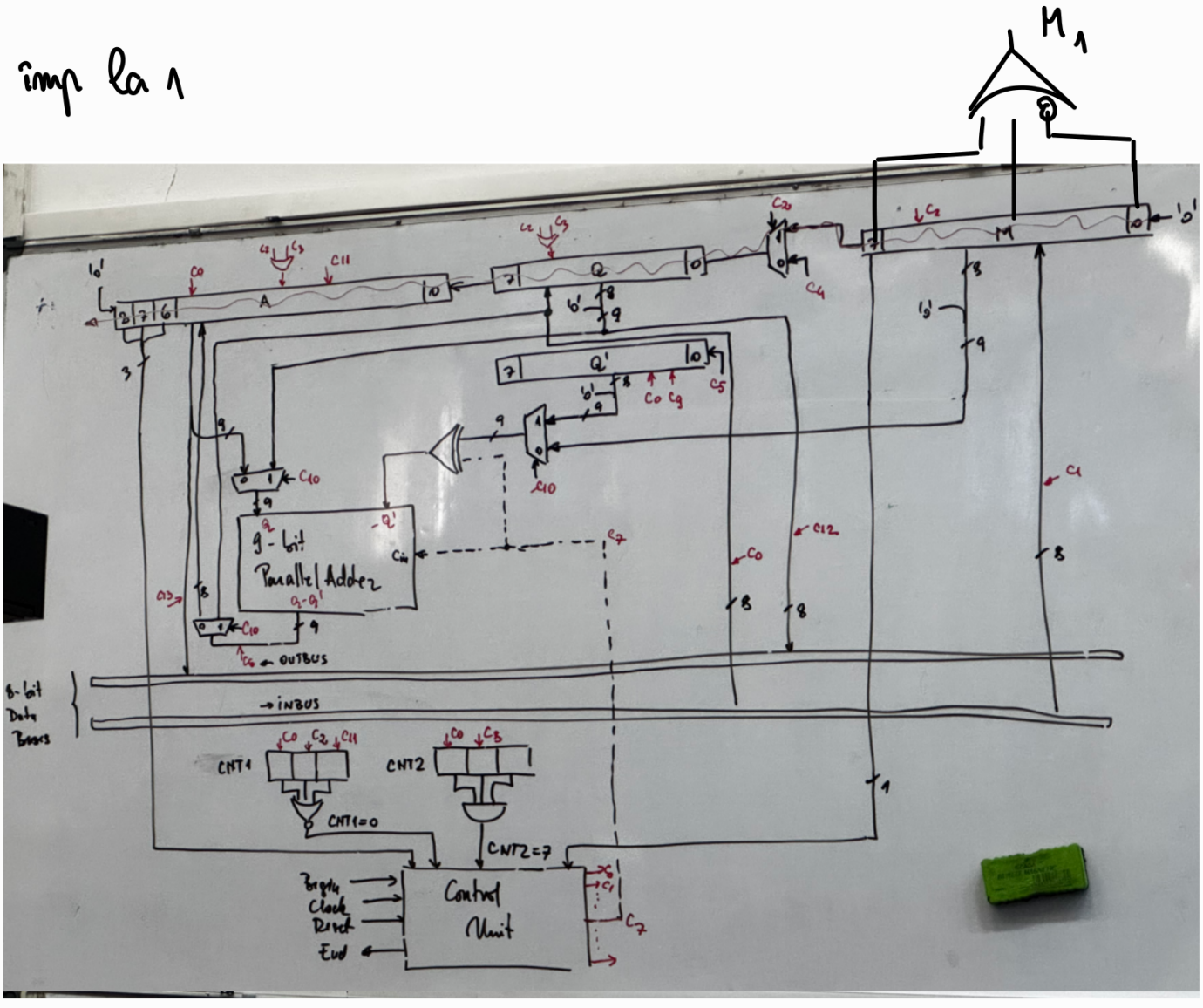


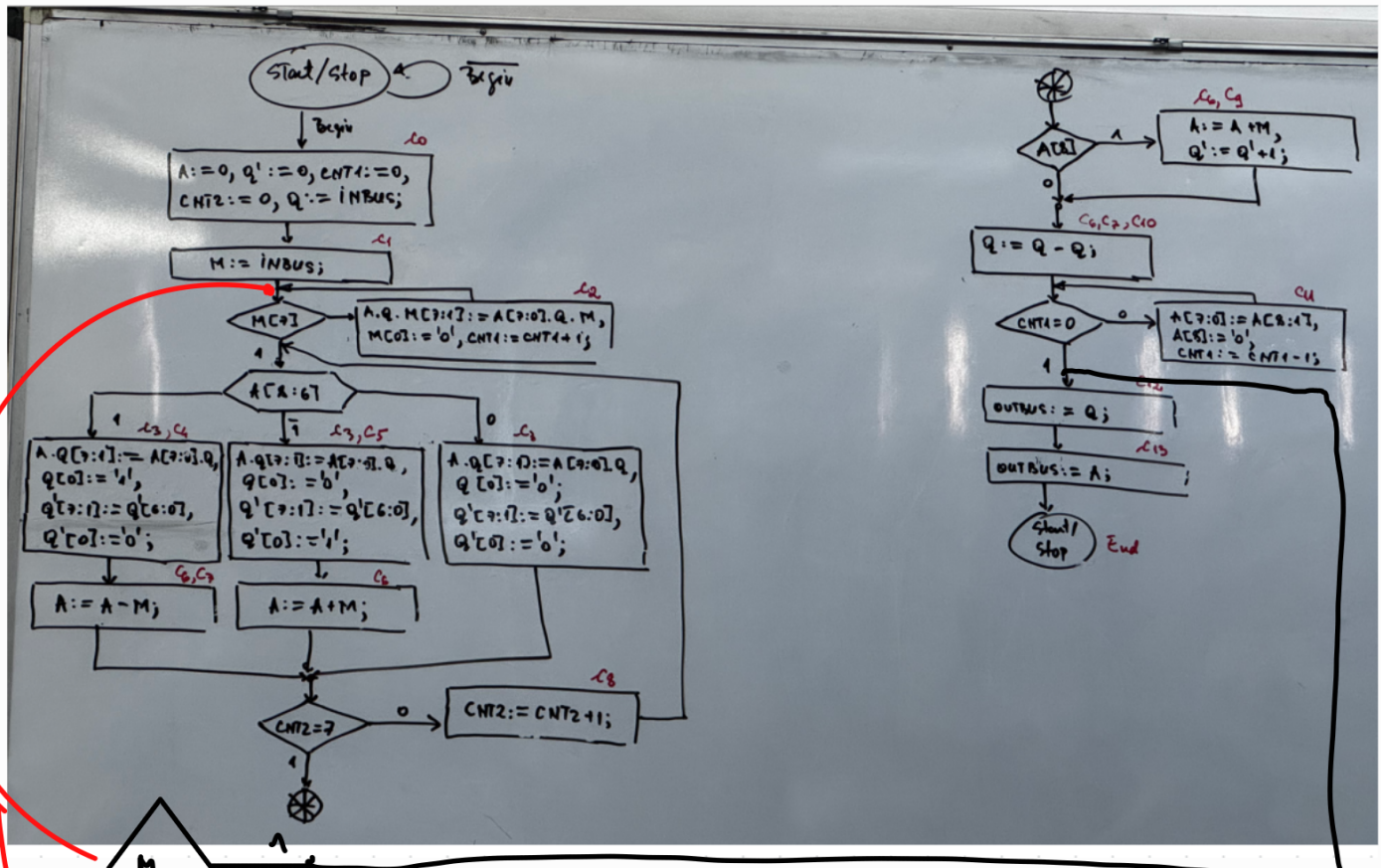

A
Q
Q'
M
9-bit Parallel Adder
CNT1
CNT2
CNT1=0
CNT2=7
Control Unit
Begin
Clock
Reset
End
OUTBUS
INBUS
8-bit Data Buses


Start/Stop
Begin
A:=0, Q':=0, CNT1:=0, CNT2:=0, Q':=INBUS;
M:=INBUS;
M[7]
A.Q.M[7:1]:=A[7:0].Q.M, M[0]:='0', CNT1:=CNT1+1;
A[8:6]
A.Q[7:1]:=A[7:0].Q, Q[0]:='1', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A.Q[7:1]:=A[7:0].Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='1';
A.Q[7:0]:=A[7:0].Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A:=A-M;
A:=A+M;
CNT2=7
CNT2:=CNT2+1;
A[8]
A:=A+M, Q':=Q'+1;
Q:=Q-Q';
CNT1=0
A[7:0]:=A[8:1], A[8]:='0', CNT1:=CNT1-1;
OUTBUS:=Q;
OUTBUS:=A;
Start/Stop
End


START/
STOP
Begin
Begin
c0
A:=0, CNT1:=0, CNT2:=0,
Q:=INBUS, Q':=0;
c1
M:=INBUS;
c2
M[7]
0
A[8:0].Q.M[7:1]:=A[7:0].Q.M[7:0],
M[0]:='0', CNT2:=CNT2+1;
1
A[8:6]
000
111
001
010
011
100
101
110
c3
A[8:0].Q[7:1]:=A[8:0].Q,
Q[0]:='0', Q'[7:1]:=Q'[6:0],
Q'[0]:='0';
c3, c4
A[8:0].Q[7:1]:=A[7:0].Q,
Q[0]:='1', Q'[7:1]:=Q'[6:0],
Q'[0]:='0';
c3, c5
A[8:0].Q[7:1]:=A[7:0].Q,
Q[0]:='0', Q'[7:1]:=Q'[6:0],
Q'[0]:='1';
c6, c7
A:=A-M;
c4
A:=A+M;
c8
CNT1:=CNT1+1;
CNT1=7
0
1
c9, c9
A:=A+M,
Q':=Q'+A;
A[8]
1
0
c10
A[7:0]:=A[8:1], A[8]:='0',
CNT2:=CNT2-1;
CNT2=0
1
c11
Q:=Q-Q';
c12
OUTBUS:=Q;
c13
OUTBUS:=A;
End
Start/
Stop

imp la 1


A
Q
Q'
M
9-bit Parallel Adder
CNT1
CNT2
CNT1=0
CNT2=7
Control Unit
Begin
Clock
Reset
End
OUTBUS
INBUS
8-bit Data Buses


Start/Stop
Begin
A:=0, Q':=0, CNT1:=0, CNT2:=0, Q:=INBUS;
M:=INBUS;
M[7]
A.Q.M[7:1]:=A[7:0].Q.M, M[0]:='0', CNT1:=CNT1+1;
A[8:6]
A.Q[7:1]:=A[7:0].Q, Q[0]:='1', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A.Q[7:1]:=A[7:0].Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='1';
A.Q[7:0]:=A[7:0].Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A:=A-M;
A:=A+M;
CNT2:=CNT2+1;
CNT2=7
A[8]
A:=A+M, Q':=Q'+1;
Q:=Q-Q';
CNT1=0
A[7:0]:=A[8:1], A[8]:='0', CNT1:=CNT1-1;
OUTBUS:=Q;
OUTBUS:=A;
Start/Stop
End
M1
1
0

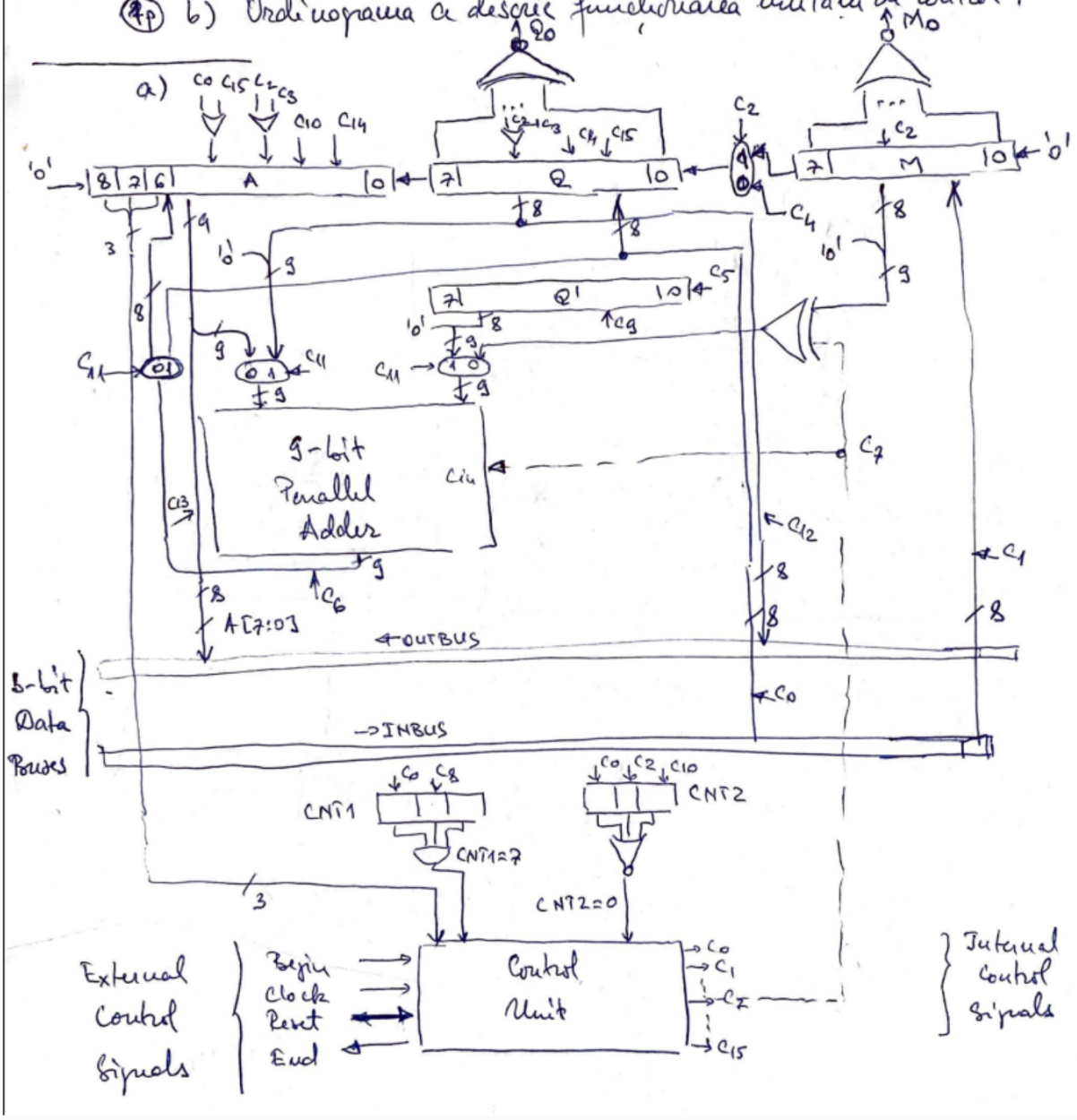
A2
B1

2p (A) Să se completeze descrierea unui împărțitor SRT radix 2, astfel încât să se trateze cazurile speciale de împărțire la zero (când atât câtul cât și restul sunt "1...1") și împărțirea lui zero. Se cer

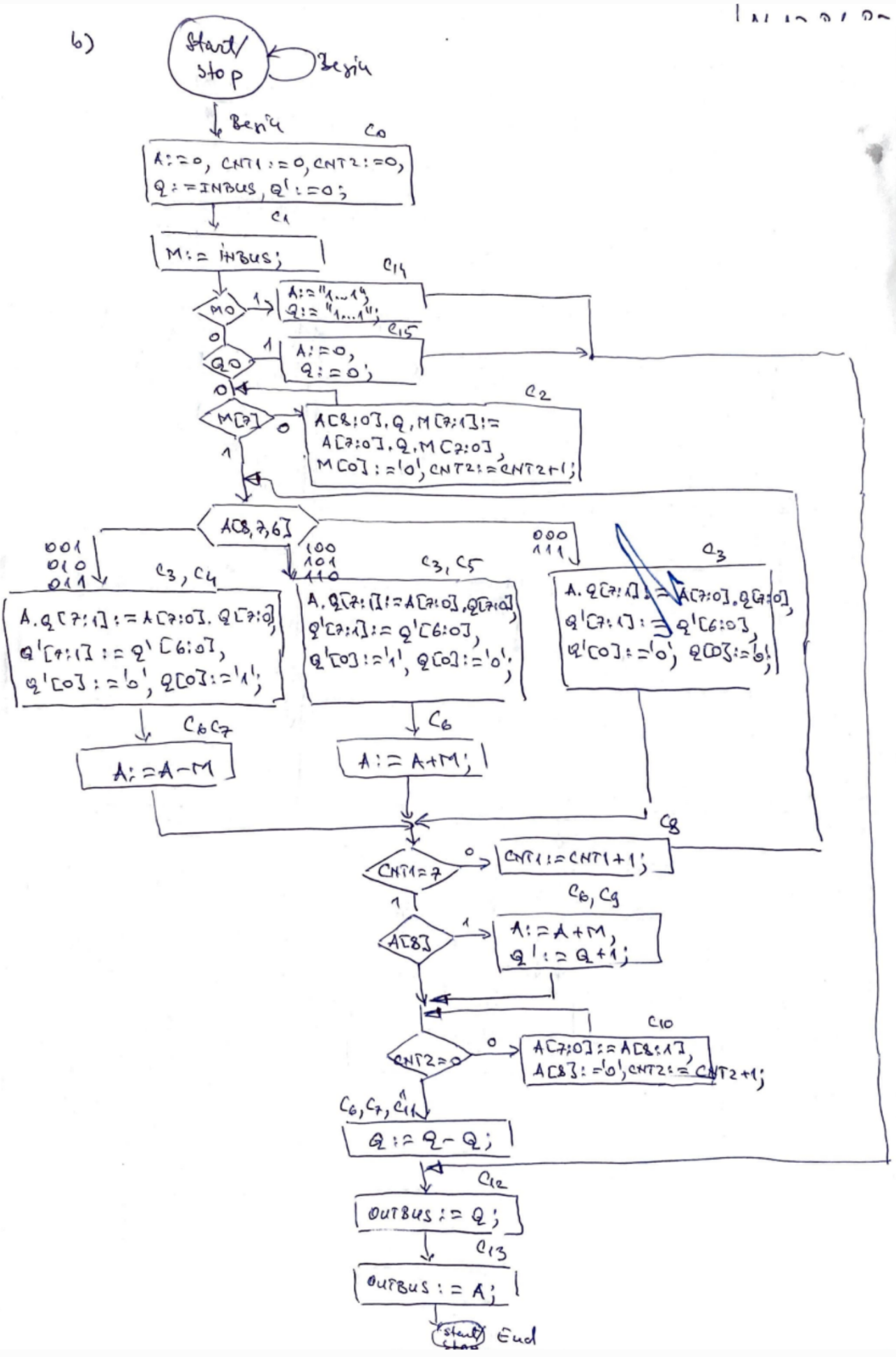
(1p) a) Schema bloc hardware + semnale de control.

(4p) b) Ordinograma ce descrie funcționarea unității de control.

a)



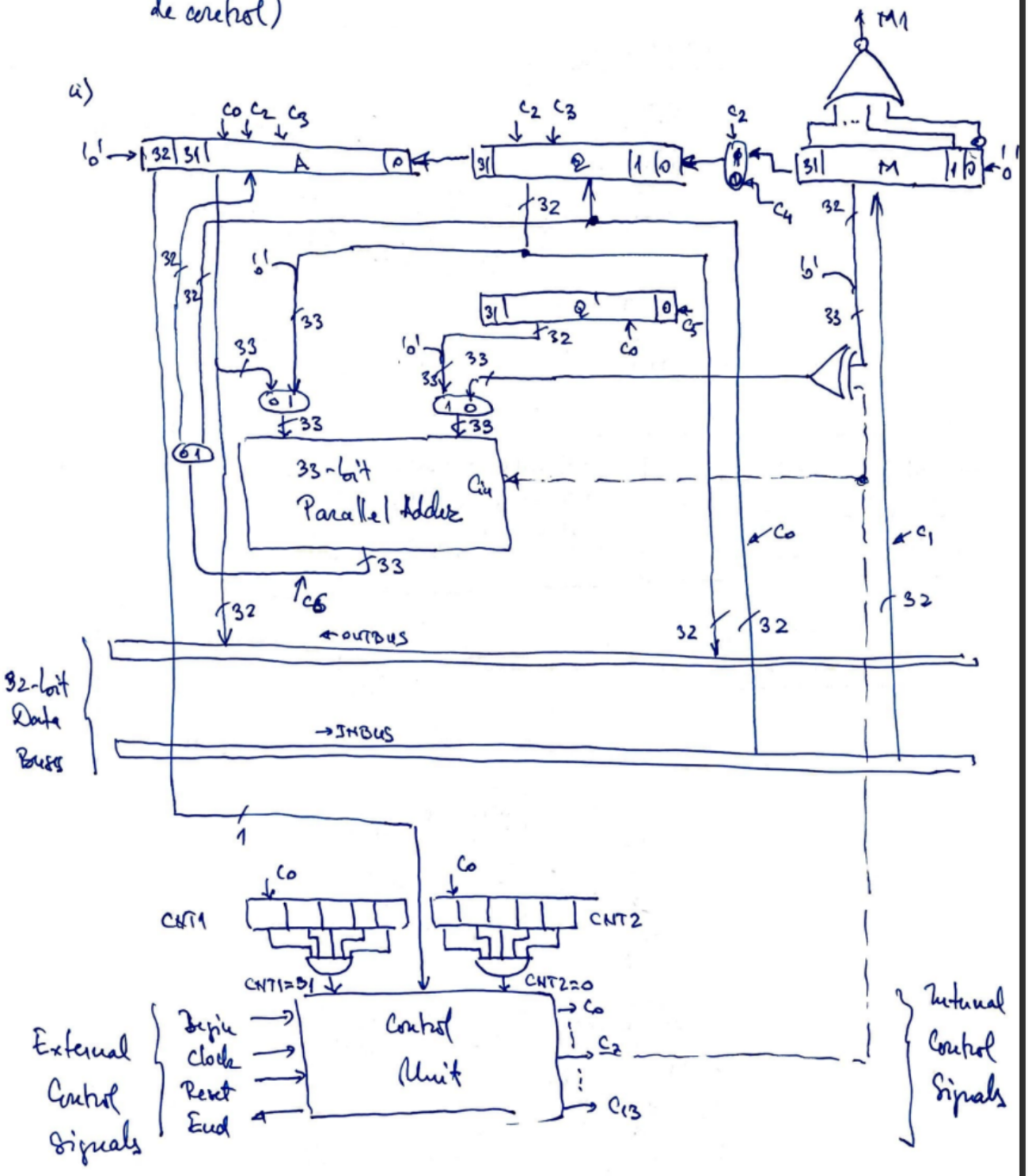
b)


Start/ stop
Beriš
Beriš
C0
A:=0, CNT1:=0, CNT2:=0,
Q:=INBUS, Q':=0;
C1
M:=INBUS;
M0
C14
A:="1...1",
Q:="1...1";
C15
Q0
A:=0,
Q:=0;
C2
M[7]
A[8:0], Q, M[7:1]:=
A[7:0], Q, M[7:0],
M[0]:='0', CNT2:=CNT2+1;
A[8,7,6]
001
010
011
C3, C4
A, Q[7:1]:=A[7:0], Q[7:0],
Q'[7:1]:=Q'[6:0],
Q'[0]:='0', Q[0]:='1';
100
101
110
C3, C5
A, Q[7:1]:=A[7:0], Q[7:0],
Q'[7:1]:=Q'[6:0],
Q'[0]:='1', Q[0]:='0';
000
111
C3
A, Q[7:1]:=A[7:0], Q[7:0],
Q'[7:1]:=Q'[6:0],
Q'[0]:='0', Q[0]:='0';
C6 C7
A:=A-M
C6
A:=A+M;
C8
CNT1=7
CNT1:=CNT1+1;
C8, C9
A[8]
A:=A+M,
Q':=Q+1;
C10
CNT2=0
A[7:0]:=A[8:1],
A[8]:='0', CNT2:=CNT2+1;
C6, C7, C11
Q:=Q-Q;
C12
OUTBUS:=Q;
C13
OUTBUS:=A;
stavi stop
End

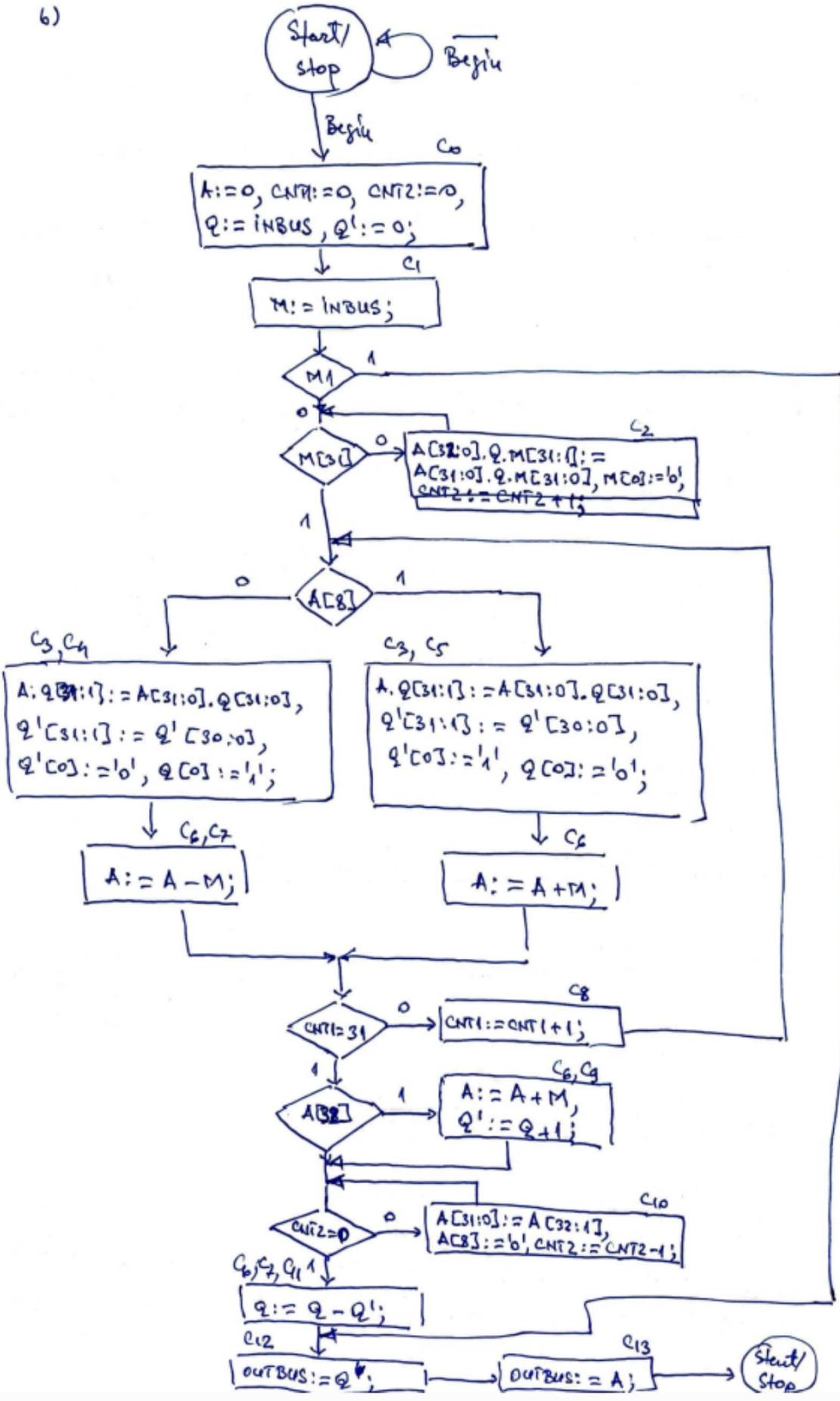
① Să se modifice dispozitivul de împărțire SRT radix-2 astfel încât acesta
② să împartă numere pe 32 de biți fără a genera cifra câtului 0 (doar 1 și $\bar{1}$) și să trateze în mod distinct situația de împărțire la 1, de ex:

a) Schema bloc hardware folosind platforma Hayes + semnalele de control

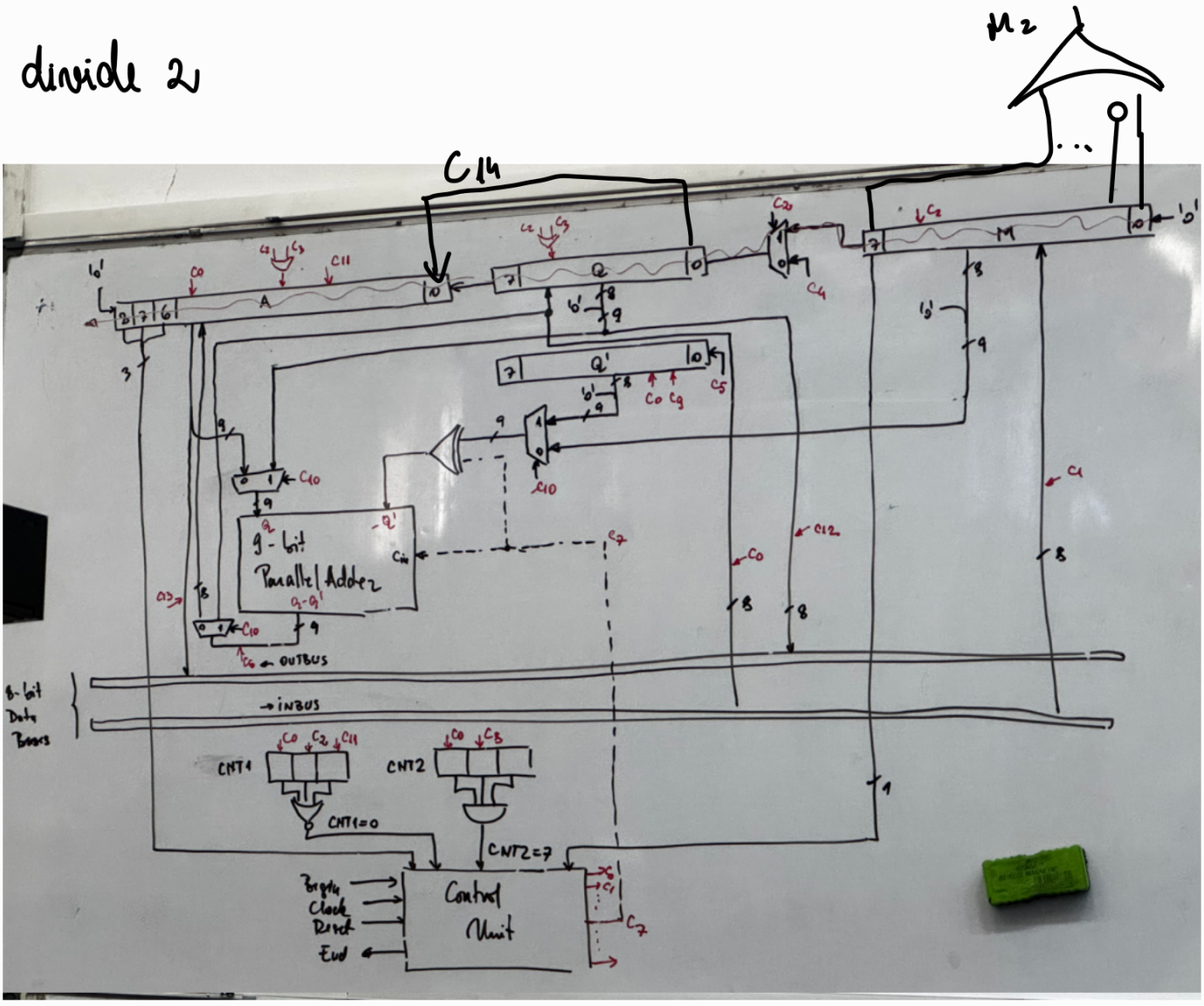
b) Ordinograma ce descrie funcționarea unității de control (+ semnalele de control)

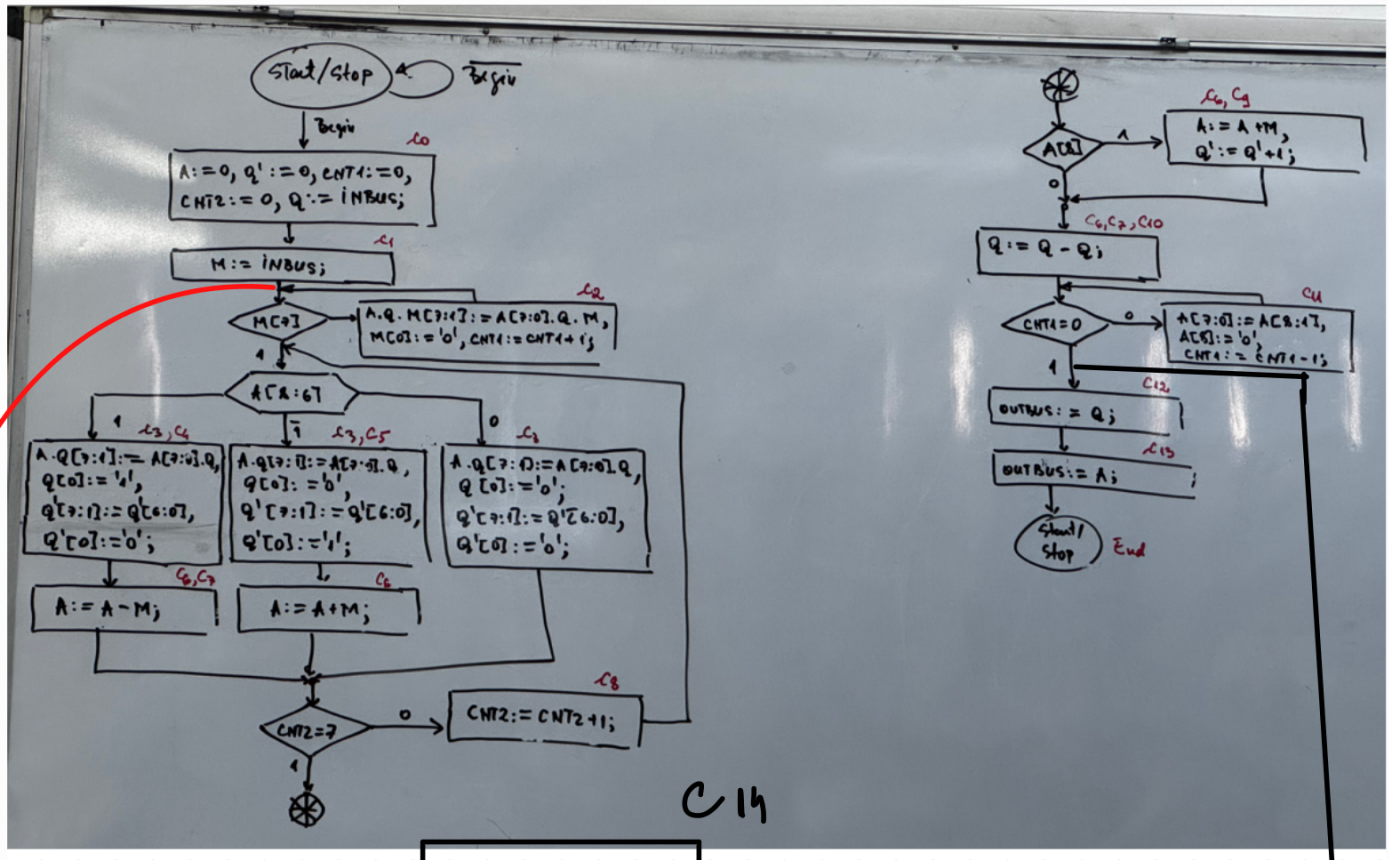


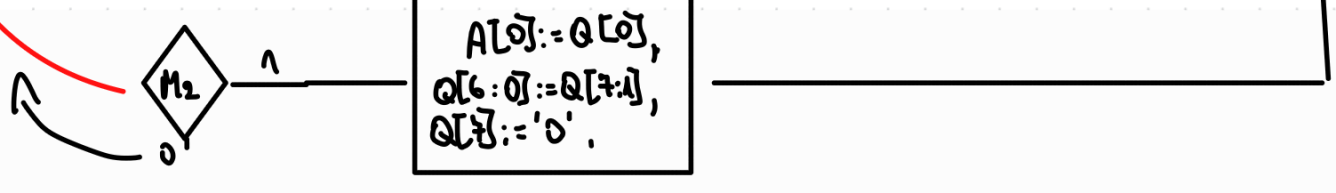
6)


Start/ stop
Begin
Begin
C0
A:=0, CNT1:=0, CNT2:=0,
Q:= INBUS, Q':=0;
C1
M:= INBUS;
M1
1
0
M[3]
0
C2
A[32:0].Q.M[31:1]:=
A[31:0].Q.M[31:0], M[0]:='0',
CNT2:=CNT2+1;
1
A[8]
0
1
C3, C4
A.Q[31:1]:=A[31:0].Q[31:0],
Q'[31:1]:=Q'[30:0],
Q'[0]:='0', Q[0]:='1';
C6, C7
A:=A-M;
C3, C5
A.Q[31:1]:=A[31:0].Q[31:0],
Q'[31:1]:=Q'[30:0],
Q'[0]:='1', Q[0]:='0';
C6
A:=A+M;
CNT1=31
0
C8
CNT1:=CNT1+1;
1
A[32]
1
C6, C9
A:=A+M,
Q':=Q+1;
CNT2=0
0
C10
A[31:0]:=A[32:1],
A[8]:='0', CNT2:=CNT2-1;
C6, C7, C11
1
Q:=Q-Q';
C12
OUTBUS:=Q';
C13
OUTBUS:=A;
Start/ Stop

divide 2


M2
C14
A
Q
Q'
M
9-bit Parallel Adder
CNT1
CNT2
CNT1=0
CNT2=7
Control Unit
Begin
Clock
Reset
End
INBUS
OUTBUS
8-bit Data Buses


Start/Stop
Begin
A:=0, Q':=0, CNT1:=0, CNT2:=0, Q:=INBUS;
M:=INBUS;
MC?1
A.Q.MC[7:1]:=A[7:0].Q.M, MC[0]:='0', CNT1:=CNT1+1;
A[8:6]
A.Q[7:1]:=A[7:0].Q, Q[0]:='1', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A.Q[7:1]:=A[7:0].Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='1';
A.Q[7:0]:=A[7:0]Q, Q[0]:='0', Q'[7:1]:=Q'[6:0], Q'[0]:='0';
A:=A-M;
A:=A+M;
CNT2:=CNT2+1;
CNT2=7
A[0]
A:=A+M, Q':=Q'+1;
Q:=Q-Q';
CNT1=0
A[7:0]:=AC8:1], A[8]:='0', CNT1:=CNT1-1;
OUTBUS:=Q;
OUTBUS:=A;
Start/Stop
End
C14


M2
1
0
A[0]:=Q[0],
Q[6:0]:=Q[7:1],
Q[7]:='0'.